BGIN Block #10 Meeting Report

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Transparency of dApps and Sound DeFi

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The Global Network for Blockchain Stakeholders™

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Block#10 Day2_6

Keynote Session Report: Transparency of dApps and Sound DeFi

Date: Monday, March 4, 2024

Location: Tokyo

Duration: Approx. 90 minutes

Summary

- The discussion begins with concerns about the soundness of DeFi systems, highlighting issues such as vulnerabilities, social engineering, operational resilience, and liquidity problems. The International Organization for Securities Commissions' (IOSCO) recent paper on DeFi, focusing on data gaps, is mentioned as a critical area of discussion. The workshop is introduced, aiming to explore transparency in DeFi applications, the availability and usability of data, reg tech capabilities, and soundness implications to drive responsible development in DeFi.

- The discussants discuss their involvement in risk management and compliance for blockchain networks, the necessity of transparent data for financial institutions, and the importance of understanding the risks in participating in DeFi ecosystems. The discussion also touches on the regulatory aspects and the challenges and needs in the DeFi space from a risk management perspective.

- The conversation shifts towards the specifics of transparency in DeFi, emphasizing the need for actionable and meaningful on-chain data, the challenges in risk management frameworks, and the vulnerabilities of DeFi services to illicit activities and operational risks. The importance of understanding the ecosystem, the stakeholders involved, and the continuous need for refined and robust data to assess risks properly is highlighted.

- The discussion delves into the practical aspects of ensuring transparency and soundness in DeFi. The importance of real-time data analysis, AI as a tool for risk assessment, and the need for a structured approach to manage the emerging risks in DeFi are discussed. The conversation also touches on the differences between traditional finance and DeFi in terms of risk management and the challenges posed by the fast-paced and complex nature of DeFi.

- The focus shifts to the challenges of establishing soundness in DeFi, considering the unique features of DeFi such as decentralization and the absence of clear accountability. The conversation explores potential strategies to improve soundness, including code

audits, partnerships, and regulatory frameworks. The importance of industry-driven standards and the role of education in enhancing the understanding and management of risks in DeFi are also discussed.

- The conversation concludes with reflections on the need for a more comprehensive and proactive approach to ensuring soundness in DeFi. The potential of self-regulatory organizations, self-insurance models, and the role of traditional financial institutions in shaping the risk frameworks for DeFi are explored. The discussion emphasizes the importance of collaboration, innovation, and adaptability in addressing the challenges and risks associated with DeFi.

- Each segment presents a deep dive into different aspects of DeFi's soundness and transparency, highlighting the complexities, challenges, and potential solutions to ensure a responsible and secure development of DeFi ecosystems.

- The discussion explores the concept of soundness within DeFi, considering computer science and financial perspectives. There's a focus on whether a unified definition of soundness can be established for financial products within DeFi and the importance of customer understanding and tolerance of risks. The conversation also touches on the traditional finance sector's approach to managing risks through controls and the need for similar meticulousness in DeFi.

- The dialogue shifts towards practical steps to improve soundness in DeFi, highlighting the importance of education across the ecosystem, including users, regulators, and financial institutions. There's a discussion on the challenges and necessity of continuous evaluation and risk assessment in real-time, leveraging AI for risk management, and the role of insurance in DeFi to protect against losses.

- Participants discuss the need for sustainable business models in the blockchain industry to ensure soundness and the challenges of short-term, high-risk business approaches. The importance of audits and third-party checks for DeFi projects is emphasized, alongside the challenges of securing insurance for DeFi protocols due to the inherent risks and the nascent nature of blockchain insurance companies.

- The conversation concludes with ideas for advancing soundness in DeFi, such as the development of standards and frameworks by industry groups like BGIN. There's a call for pinpointing life-or-death risks in DeFi, particularly focusing on cybersecurity and code vulnerabilities, and the role of self-regulatory organizations and insurance in fostering a safer DeFi ecosystem.

Transcript (anonymized):

01:59:02
It's truly a pleasure. In discussing the realm of decentralized finance (DeFi), it's become clear that there have been significant challenges related to the robustness of these platforms, stemming from vulnerabilities in smart contracts, both in terms of their programming language and actual code. Additionally, there's been a notable issue with social engineering, targeting individuals involved in these platforms, along with other concerns such as operational resilience, liquidity, and the prevalence of fraud, indicating deep-seated problems with the overall integrity of these ecosystems. The focus today is on the critical issue of transparency, a topic thoroughly examined by various international standard-setting bodies, highlighting the need for scrutiny in this area.

02:00:01
Recently, a notable publication on DeFi highlighted key challenges, including significant data gaps, providing an intriguing point for discussion. This paper is essential for understanding DeFi's integration into regulatory frameworks primarily based on disclosure, such as those governing capital markets and consumer protection. The discussion will delve into the kinds of information currently available, its underutilization, and how enhancing information access can improve oversight capabilities and assist commercial entities in comprehending the risks associated with DeFi ecosystems. This workshop is conducted under the Chatham House Rule, allowing for open discussion while maintaining confidentiality of the information source.

02:00:47
Today's session features distinguished speakers, including Fletcher Roberts from a notable tech company, Cheng Yi from a leading blockchain analytics firm, stepping in for Caroline Malcolm, and Nikos Andriko Jianopoupoulos, the CEO of a prominent risk management company. The discussion will pivot around the vital need for transparency in DeFi applications, the accessibility and practical use of data, and the role of regulatory technology in leveraging this data to foster sound development within DeFi. The speakers will share their insights on transparency, the robustness of DeFi within decentralized applications (DApps), and their perspectives on essential areas for improvement to enhance transparency and integrity in DApps.

02:01:46
Greetings to all. I'm the founder and CEO of a company specializing in risk management and compliance for blockchain networks, offering a unique approach that supports financial institutions in integrating new digital assets and services. Our platform enables these institutions to establish comprehensive risk frameworks, comply with regulatory requirements, and monitor metrics in real time, thereby facilitating a seamless transition into the digital asset space and ensuring they are equipped with the necessary compliance tools.

02:02:35
I'm delighted to contribute to discussions with global regulators through my involvement with the Commodity Futures Trading Commission (CFTC), highlighting the collaborative efforts required to address the evolving landscape of DeFi and DApps. The focus is on understanding the perspectives of financial institutions regarding DeFi and outlining future directions for the industry.

02:02:53
Thank you for that introduction. I'd like to acknowledge the valuable contributions made through the joint work on the CFTC's technology advisory committee, which recently published an extensive report on DeFi. This report addresses various risks, including those related to the integrity of these ecosystems and the challenges associated with transparency. We will explore some of the proposed solutions and discuss recent publications aimed at tackling these issues, emphasizing the importance of transparent and sound practices in DeFi and DApps.

02:03:45
Hello, everyone. I'm here on behalf of a prominent blockchain data analytics firm, stepping in for a colleague. Our company focuses on analyzing cryptocurrency transactions both at an individual and macro level, aiding law enforcement and businesses in risk management and compliance. My background includes extensive experience in financial regulation, supervision, and development, having spent a significant time at a major financial regulatory authority. The insights and experiences I bring from these roles fuel my enthusiasm for today's discussions on the intricate dynamics of transparency and soundness in the DeFi space.

02:04:50
The discussion today needs to pivot towards two critical areas: transparency and the overall robustness of DeFi platforms. From an on-chain data perspective, the visibility of DeFi transactions offers a promising avenue for enhancing transparency. However, the challenge lies in making this vast amount of data actionable and meaningful. An area of concern is the activities and risk management practices that occur off-chain, which remain obscure and are areas where we should strive for greater clarity and openness.

02:05:40
Moving onto the aspect of soundness, there are two primary risks that need addressing. First is the susceptibility of DeFi platforms to exploitation by malicious actors for purposes like money laundering. The second significant risk is the exposure to cyber threats and operational vulnerabilities, such as hacks and thefts. The presence of experts from leading regulatory technology firms in today's discussion brings a wealth of knowledge focused on mitigating these risks and enhancing the transparency and integrity of DeFi platforms.

02:06:36
Proceeding with our agenda, we encourage an interactive and engaging discussion among all participants. The focus will be on critically examining the issues surrounding data availability, analysis, utilization, compliance, and integration within the DeFi ecosystem. Our aim is to dissect and understand the nuances of transparency and how it serves as a foundation for ensuring the soundness and reliability of DeFi services.

02:07:21
The perspective on transparency varies widely among different stakeholders, including financial institutions, regulatory bodies, consumers, and law enforcement agencies. The meaning and necessity of transparency can differ based on the context and the specific requirements of each stakeholder group. Today, we seek to explore these varied perspectives and understand the multifaceted nature of transparency in the DeFi ecosystem.

02:07:44
Focusing on the experiences from the financial sector, it's clear that the engagement with DeFi and decentralized applications (DApps) is primarily with the larger, more established platforms, particularly lending protocols. The level of transparency and information availability within these platforms significantly influences their adoption and integration by financial institutions, highlighting the need for clear and accessible data to facilitate risk management and compliance within the broader financial ecosystem.

02:08:32
From the perspective of financial institutions, transparency is defined by the availability of sufficient data that allows these institutions to confidently provide custody services, onboard assets, and offer additional services. This level of transparency is critical, albeit only a few financial institutions engage deeply with DeFi tokens. The process of a DeFi project or DApp being listed on an exchange, such as Coinbase, serves as a practical example of this transparency in action, where the exchange's risk framework and listing requirements play a significant role in determining the level of exposure and acceptance a DeFi token might receive.

02:09:23
The approach taken by exchanges like Coinbase, which may assume greater risks compared to others, especially in cases like Bitcoin ETFs, underscores the varying standards of transparency and risk tolerance within the market. The fundamental aspect of transparency at this stage involves building a community around a project, ensuring sufficient confidence for token distribution and community contributions, based on a shared belief in the project's transparency and vision.

02:10:05

A crucial realization for projects aiming for institutional adoption is the necessity of making relevant data available. This includes detailed scrutiny of GitHub repositories to assess the structure, quality, and security of the code, alongside regular audits. Such practices are integral to financial institutions' risk assessment frameworks, where questions about code structure, update frequency, and audit history are paramount.

02:10:48
Beyond technical analysis, social media sentiment and engagement across platforms like Discord and Twitter provide additional layers of insight into the project's reception and community dynamics. The analysis extends to the structural integrity and dependencies of smart contracts, which are central to the functionality and security of DApps. Information on project founders, key personnel, and investors further enriches this data pool, offering a comprehensive view of the project's credibility and governance structure.

02:11:31
Governance mechanisms, community proposals, and participation rates are also critical data points, accessible through various platforms and forums associated with the project. The depth and organization of this data tend to correlate with the project's size and its aspirations for institutional adoption, highlighting a trend towards more structured and comprehensive data availability for larger protocols.

02:11:56
The discussion then shifts towards the nuances of transparency within these ecosystems, questioning the balance between public transparency and selective disclosure to specific counterparties. This raises important considerations about the nature of information sharing in DeFi, probing the extent to which public access to data is necessary versus more restricted forms of transparency tailored to business or individual needs.

02:12:25
Currently, the transparency in decentralized applications (DApps) largely caters to the community and crypto investors, making most information publicly accessible. However, as regulatory frameworks evolve, certain data may become exclusive to regulatory requirements, introducing a different level of transparency classification that isn't public. This shift hasn't fully manifested in financial institutions' requirements yet, but it's an area to watch as the landscape evolves. The essential question arises about the specifics of transparency: what needs to be transparent and to whom? The answers vary depending on the stakeholder, whether it's financial institutions, regulators, exchanges during initial coin offerings (ICOs), or the community during token launches, with the transparency requirements intensifying as one moves towards institutional adoption.

02:13:23
In practice, transparency takes on different forms for various stakeholders. For financial institutions and regulators, a higher level of transparency is required compared to what is needed for community engagement during token distribution. Interestingly, financial institutions often demand more transparency than regulators, attributed to the latter's limited understanding of the space. This highlights a layered approach to transparency, where the requirements intensify as DApps move towards institutional engagement, with each stage demanding a specific focus on what needs to be transparent and to whom.

02:14:01
The conversation then shifts to another perspective on transparency, emphasizing its contextual nature. Depending on the stakeholder—be it commercial entities, consumers, regulatory bodies, or law enforcement—the definition and necessity of transparency can vary significantly. The pivotal question is whether all information needs to be public for a system to be considered transparent, especially within the DeFi ecosystem. Transparency must serve a purpose, which changes with the stakeholder in question. For law enforcement, the focus is on tracking illicit activities on-chain, where specialized tools are essential for analyzing and making sense of complex smart contract interactions.

02:15:00
For law enforcement, transparency is crucial for tracing illicit activities within the DeFi space, with the emphasis on the ability to analyze on-chain transactions. This level of transparency, while technically available, requires specialized analytical tools to interpret complex interactions, indicating a barrier to entry for general public access. Thus, while the information is accessible, its utility is contingent upon the availability and use of these specialized tools.

02:15:46
The tools in question are often utilized by those involved in law enforcement, as they have a vested interest in acquiring such tools. This does enable a degree of transparency, yet it is also accompanied by a significant barrier to entry, stemming from the reality that not all individuals require access to this extensive information. This outlines the concept of transparency within the realm of law enforcement. Shifting focus to another group of stakeholders, such as central banks and regulatory bodies, transparency for them hinges on their understanding of risk locations and the identification of accountable parties. The clarity in these scenarios can sometimes be less distinct.

02:16:25
For central banks with concerns about prominent decentralized applications, the interconnections between decentralized finance (DeFi) and centralized services, and the extent of domestic user engagement with these platforms, the available information is plentiful but not always as detailed

or refined as the traditional financial risk monitoring frameworks might demand. This is partly attributed to the rapid development and relative novelty of the ecosystem. Despite being one of the more established entities in blockchain analytics, having not yet reached a decade in operation, the challenge remains in sifting through the vast information to pinpoint relevance and present it in an easily digestible format.

02:17:24
For instance, while direct observation of rehypothecation on the blockchain may not be feasible, it is possible to trace the movements between, say, a lending protocol and a staking protocol, thereby visualizing a significant portion of the activity. However, this necessitates posing the correct questions, and it falls upon data providers to package this information in a format that is both comprehensible and actionable. This represents the second tier of transparency-related challenges. Returning to the perspective of regulators, the recurring question is the clarity over who within a DeFi service holds accountability or exerts influence over operations, including cyber resilience, fee structures, and decision-making processes.

02:18:20
Questions arise about whether decision-makers are token holders, voters, backend developers, or frontend supporters facilitating user access to decentralized applications (DApps). This area likely experiences the lowest degree of transparency across the board, given the variability from one DApp to another and over time. Considering the recommendations from regulatory bodies on DeFi regulation for market integrity and consumer protection, a broad spectrum of potential accountable entities is identified. However, the dynamic nature of these entities across different services introduces inefficiencies in pinpointing accountability, prompting a reassessment of transparency objectives and possible approaches to narrowing down areas of concern.

02:19:44
The emphasis on understanding and managing risk is crucial for both regulators and law enforcement agencies. This involves not only probing into high-risk or clearly illicit activities but also identifying systemic or stability risks within the ecosystem. The current state of transparency within decentralized applications is a subject of inquiry, alongside the developments in layer two solutions and other components not directly on-chain, questioning their level of transparency.

02:20:29
To aid in this discussion, reference is made to an annex of a consultation report by a regulatory body on DeFi recommendations, which delineates useful metrics for evaluating DeFi protocols. This annex, accessible through a shared link, covers a comprehensive range of information, some of which is publicly available on DApps. The report further delves into the metrics essential for

analyzing smart contracts, protocols, blockchain platforms, and the broader DeFi market, highlighting the perceived gaps in financial reporting amidst the vast data landscape.

02:21:12
The discourse touches on the challenges related to reporting—whether it's the distillation of information into a useful format, the non-disclosure of crucial data not available to the public that regulators and law enforcement believe is necessary, or the overall scarcity of comprehensive market data. These issues underscore the broader dilemma of transparency within the DeFi sphere, particularly the delineation of accountability lines. This context sets the stage for further discussion on the transparency of decentralized applications (DApps) and the visibility of information across the DeFi ecosystem.

02:21:53
Addressing the topic of transparency, it's acknowledged that the field of blockchain analytics, including companies specializing in Know Your Customer (KYC) and Anti-Money Laundering (AML) protocols, has seen significant growth over the past decade. The availability of data concerning risks beyond KYC and AML is vast, with the KYC/AML sector reaching a level of maturity that allows for the integration and labeling of data from sanctioned wallets, among other advancements. The pursuit of even more detailed data continues, with efforts to procure information from existing providers and generate new datasets.

02:22:41
It's suggested that achieving the granularity of data required by financial institutions might span another decade. The notion of transparency is complex, varying not only by the recipient but also by the level and type of transparency needed. The requirements of financial institutions for transparency fundamentally differ from those of the web3 communities. For instance, in web3, transparency might be achieved through platforms like Discord and the distribution of tokens, whereas financial institutions might demand detailed financial statements, shareholder information, and legal compliance, akin to the standards upheld by entities like Circle with USDC, which includes audited financial statements.

02:23:23
Looking at DApps through the lens of the web3 community post-crypto winter suggests a cycle of renewed transparency aimed at the next potential token offerings. The future of which DApps will evolve and meet regulatory and financial institution standards remains uncertain. However, in regions with active community participation in apps, such as APAC, transparency can lead to increased user engagement. Significant user investment might necessitate custodial solutions, prompting financial institutions interested in this market segment to consider how to incorporate these tokens into their risk and regulatory frameworks.

02:24:15
The discussion concludes with the anticipation that progressive custodians might explore ways to accommodate the holding of tokens within their services, balancing the business potential against regulatory constraints and risk management considerations. This involves a collaborative effort among risk, product, and business teams to navigate the evolving landscape of DeFi and DApps, seeking a harmonious integration of new digital assets into traditional financial systems.

02:24:15
So to the extent that they are super transparent, they get a lot of users. Those users, if they have significant exposure, they would need at some point some custodian for those assets. So then that propagates to financial institutions that say, if we want that part of the business, we might want to figure out a way of being able to kind of hold those tokens and have our risk frameworks accommodate those. So I think you might see some of the more progressive custodians thinking about how can we do that? Is there enough data and kind of swinging between the business case and the regulatory framework and kind of between the risk people and kind of the product people and the business people kind of trying to figure out what's going on?

02:24:56
I totally agree with you, that seeing the evolution in this space, the insurance market is a great example of where any sector where hedging risk is important. An insurance ecosystem normally arises out of market ecosystem. It's a great business model. If they understand how to assess risk, it will develop. Insurance is not a realistic thing right now in crypto. And the only insurance companies that I do see are really, for the most part, insurance for cyber insurance. And even cyber insurance as a sector has a lot of maturing to do. It has had a lot of maturing that it's done over the past six years, but there's a lot that has to.

02:25:38
I think the fact that there isn't yet a stronger insurance market willing to play in crypto demonstrates that there's still an issue with availability of information to help people feel confident in their ability to assess risk in the crypto ecosystems.

02:26:04
I wanted to offer different definition of transparency, if I may, and that comes from the perspective. What happens when things go wrong. And coming out of the traditional financial markets, we had a flash crash in May 2010, which took the regulators six months plus to actually figure out what happened in the market and ultimately led to the consolidated audit trail. So ten years later, I think they're actually getting to implementing it now. And the transparency that you need to figure out what actually happened in the market when something goes wrong, it's totally different from all the metrics you had which exist in traditional financial markets as well. What's the trading volume, what's the high low, what are the tick moves, et cetera.

02:27:06
But can you retroactively, really reconstruct the market almost millisecond by millisecond as a snapshot is a totally different type of transparency.

02:27:18
Does the group think that DeFi is well placed to provide that, given that blockchain was made for data integrity and auditability, it is potentially a solution to that, as long as you trust the ledger, which presumably you do if this DeFi application is trusted. So potentially, the transparency could provide that, if the information that's being recorded is the relevant information to tell you what did in fact happen in the market. Because if what's happening is something behind the scenes, that's the greater indicator of the, let's say, massive market manipulation that may or may not be totally happening inside of these spaces. Can we see the indicators of it based on publicly available data, or what is the other information off chain that we need to really know what's happening behind the scenes?

02:28:06
If you want to weigh in on the state of how transparent are these systems, what is the state of the reg tech in leveraging availability of this data? Just to pick up on this point on transparency being this sort of idea of, is there enough of a data trail, information like sort of transparency, so that if something goes badly wrong, you can then go back and forensically analyze what happened, what went wrong, what were the flows, where was the manipulation, so on and so forth?

02:29:01
You are right that DeFi does offer more potential for that, provided the regtech keeps up. The regtech is trying, is working very hard to keep up.

02:29:21
I'll get to that in a minute, but I think the bigger bugbear around transparency in that context is not like X post. Can I figure out what's going on, but rather X ante? Can I figure out what's going to go wrong and if something goes wrong, how that will cascade through the system? And I think there's always an element of data availability, but I think the bigger problem is an understanding of the ecosystem and how it's linked.

02:29:59
There is this concept of an elithiometer, which is kind of like a watchlike device that will tell you the answer to any question that you want, provided you ask it in the right way and are able to interpret it in the right way. And you're very limited in the way that you can sort of want an elithiometer oracle so bad.

02:30:19
Okay, sorry.
Please keep going. I love that. No, so I think about blockchain data in the same way. It is out there. It is laid out, it is transparent. There are obvious limitations, but it can tell you a lot if you know what questions to ask. And I think that the problem where we are now with regards to all these questions around DeFi and where the risks lie in the ecosystem, there is a chunk of that centers around us not knowing what is the right question to ask. And that is a more fundamental question. And I think it comes back to, again, drilling down into transparency. So transparency, we know broad buckets, like, what is it that we're trying to get at? We're trying to get at, where is the locus of risk? But what exactly does that risk look like?

02:31:00
That's the question that we need to pinpoint as a community before we even think about whether that data is.

02:31:09
I mean, one question I have on my side is, how do you distinguish between Dapp and DeFi? Because in my, like, it's mostly DeFi when we care about things that have a financial impact, like Dapp, are you thinking about, I don't know, decentralized Twitter, decentralized Uber? These are things that I don't think apply yet to any financial institution or regulator. They mostly apply to the community at this stage, like most of the DeFi. I think some of the DeFi now is reaching some financial institutions, but still most financial institutions are only looking at bitcoin and Ethereum right now.

02:31:46
Yeah, that's a fair point. You're right that I'm sort of defaulting to DeFi as the Dapps that I'm talking about because in my current view, that is the highest risk version of the Dapps that we're seeing. And those are the ones that are being poned the most because of their practical issues with soundness. But you're right, soundness looks different for.

02:32:04
Something that both transparency and soundness. The main objective is convince a community that the project is worthwhile so that they can spend time and they can participate in the token distribution. Right. The risk of all of that lies with the individual in the web 3 casino. To the extent they believe it's a casino. To the extent they don't believe. But they take that risk. Nobody protects them. There is no centralized provider there. You take the risk based on your individual judgment of the guys on YouTube, on Twitter, on everything that you see, you make a judgment that, should I be investing my time, maybe my money that, and kind of participate. So I think

that soundness and transparency, to me when it comes to the majority of the apps, has to do with risk that individuals take in this space.

02:32:48
Yeah, I appreciate both those points.

02:32:51
And I know that there is a bigger question whether venture capital has some responsibility when they buy tokens at a certain price ahead of ICOs, and whether they should have risk management and they should have different criteria on transparency and soundness before they put the rubber stamp on those tokens and people feel excited about participating.

02:33:11
That's awesome. Besides the insurance space, the other space that generally is specifically in the market in order to successfully hedge risk is the investment community. That is another space and one that is more mature is an interesting. Where it exists in a stronger market sense, I'll say maturity in the investment community, there's a roadmap that they still have to be able to accurately assess risk. Otherwise we wouldn't have seen hundreds of millions of dollars going to fraudulent platforms.

02:33:44
I mean, they invest out of funds that are outside the US, right. They set up crypto funds as a separate vehicle so that they can buy those tokens. If you look at a 16 z or others, they have special vehicles for those purposes. So the type of regulation that applies there is a big question, because when they set the threshold for the token value ahead of the rounds, the community sees that and they participate. And in their mind, somebody else has done their due diligence, so they piggyback on that.

02:34:19
Because I'm going to close this out a little bit on transparency, only to move us into soundness. But the issue of transparency comes into our ability to assess soundness here in this chart, which is very long. Like what are the data elements that are needed for each layer? For example, blockchain transactional data and mempool data? What is available for them to see and what are the potential gaps in challenges that they see? So many of the challenges that they point to are not about absence of data. It's about the ability to use it, distill it, which I think points back to a comment about are the right questions being asked? Do the regulators even know the right question?

02:35:03

Every single question associated with what they need at the settlement layer was not listed explicitly here. That doesn't mean that they couldn't do it. I am curious if they could do it. But truly that level of specificity about what are all the specific questions that are being asked, that level of specificity is something that I haven't really seen exist in academic research. That's something that we could consider expanding upon. That's a huge amount of work for every single possible regulatory and law enforcement need. That would be useful. But going through for each layer they identify at the asset layer, they pointed out GitHub and how there's no assurance about the information that's available there and how accurate it is for smart contract layer specialized skill sets. Skill sets was listed in several locations.

02:35:55
So again, it's back to the capability of the authorities to be able to ingest the information, which can be a regtech gap and a place to make money for the regtech community. Or maybe that skill set will learn that actually the information isn't available and maybe that's an adaptation that has to happen in the ecosystem.

02:36:15
Financial institutions have a corporate VC. I suspect that when those corporate VCs of the bank start participating in those kinds of investments, they will start doing their risk management a different way that traditional crypto VCs have been doing in the past, right? In the past kind of three, four years, the VCs have been kind of split into two camps, those that didn't touch crypto and thought it was a scam. And the crypto VCs on the other side, which the discussion was very quick. Do you have a token? What's the story? If you have, what does the market cap look like and done and they finance you. So that is what propelled, I think, those kind of dApps and DeFi protocols.

02:37:01
Once they had a compelling story and they met the VC threshold of thinking that this is a token that is worth it, but that's not robust enough. I think maybe the financial institutions that are more aware of what risk management means, maybe they start investing, maybe they start educating the regulators there. I think there would be both on the regulatory side and on the finance institutions, some kind of enrichment of the risk frameworks for DeFi and DApps.

02:37:30
Yeah. Do we think that those risk management frameworks are mature enough across that space? Like if we partnered with the venture capital and private equity community, the VC?

02:37:40
No, but you'll see some of the top financial institutions have very robust risk frameworks. I'm very surprised. This is work that top consulting firms in the world have been doing for the past

three, four years. And it took years, literally, for them to be doing those. And now they have those in place. And this is the year where you see with bitcoin ETFs starting the year and ETH ETFs later on, this is the year where these risk frameworks start getting applied in the yet. I mean, DeFi is only there in terms of stablecoins and lending protocols and only in very few of the financial institutions.

02:38:12
I was just going to echo that only because I think one of the frustrating things was that the environment of tech VCs, especially this concept of, I don't think you're going to see that because I really think that industry is so unique in that they are so used to betting on worthless things.

02:38:55
They will invest money in many ventures because they're looking for one to pop at like 100 x returns. And nobody else has such low risk management, low due diligence. They're basically gamblers with titles. And so I don't think you're going to see venture capital within a regulated financial institution, I think is such a different horse of a different color that I just don't think you're going to see that evolution until these projects, it's such a different level of diligence, I don't think when people talk about where's the institutional adoption? I think it's so far for that reason that I just don't think many of these projects are going to get to a level of being able to comply with that level of diligence.

02:39:59
It is definitely 90% of the time gambling. And there's not really a culture around safe investments, especially now that markets are heating up during the bear market investments and blockchain funding was far more, let's call it smart. Whereas now, and also back in sort of other bull runs, it's been definitely high risk.

02:41:02
And so it's a very high risk space in which most people that work with blockchain companies that get funded would understand. So I think moving to enterprise blockchain, so existing companies that utilize blockchain technology and try to put some R&D budget to implement it into what they're doing far more safe and of usual going back to audibility and all those sort of core values are there, whereas it can just be with some startups. What's your token and what's the sort of the monetization of this particular DeFi protocol that you're in? Yes, unfortunately,VC's job is to make sure that it is validated and secure in as much ways as you can possible, especially on the smart contract layer and sometimes other layers.

02:42:06

Depending on market cycles, it is very high risk indeed. Let me take as an example. Right, Eigen layer, an amazing project that's doing risk taking on Ethereum. It's extremely risky, right. This project of risk taking and taking decentralization as a service from Ethereum anywhere else is extremely risky, right? Even Vitalik wrote a blog post about how he thinks there are certain risks to the Ethereum community, so much more to the financial institutions. However, they got funded last week and recent led around 100 million, 6 billion of inflow. Right now, the industry is kind of hungry for making money out of, particularly around Ethereum. It creates more and more concepts of decentralization, of lending trust. Nobody knows where the risks are and how they're buried.

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How can you really tell what are the criteria by which Eigen Layer admits a project in that and what kind of risk you're taking and how many people are monitoring that kind of risk? And then that's where the insane thing is. To the extent that Ethereum's price reflects the value of total value locked of projects like that, and that starts creating a systemic risk to financial institutions that are now providing custody on Ethereum and the ETFs and all of those things. So I think there is significant risk that we're going to see from DeFi, like propagating on the value of the native token and all the finance digital asset services that are being provided for that.

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a lot of the excesses that we've seen took place in an environment where the world is basically like awash with liquidity. Right. And there was no real incentive to be responsible with where you were placing your bets. I do think these market cycles can be instructive.

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So if going forward, we're going to be slightly tighter liquidity conditions, we're already seeing, we've seen how VC funding to the web3 space has dropped sharply. And so going forward, if money is going to be more expensive, is there going to be more incentive for these VCs, whether they're tech VCs or CVCs, to place their bets carefully with more of a risk lens? And then if that is the case, then is the challenge really just more that they don't know where the risks lie? And then how do we solve that problem.

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I think it's as simple as exchanges need to be heavily regulated. They are heavily regulated. They violate OFAC rules and they don't care. They'll pay the fine. That means totally unacceptable. I understand, like the Ethereum ethos, but if you are a US publicly listed company that you have taxpayer money from being invested in your company, you cannot be taking that to a level that is like the highest standard ever. And I think every regulator in every country should be imposing very strict rules for listing and delisting. Like NYDFs recently kind of published about delisting rules on the tokens from exchanges. Because a VC, that's what they see, right, as their exit

scenario, being listed on an exchange having wide distribution and massively selling, and that's their exit strategy.

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If you make that stricter, then they will apply those criteria earlier on in their due diligence. They will say, this project has no fate if it doesn't go through that. And then I think that the question is if VC starts seeing financial institutions being like the end exit of being listed through ETFs and through even wider distribution, that kind of the exchanges, I think those kind of strict risk management and compliance rules need to propagate backwards towards the investment community.

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Yeah, I think that's a really interesting point, moving into soundness of the system generally, that in order for DeFi applications to become more sound, you also need the CeFi entities that are writing on top of a lot of these layers to be sound. Some of these obligations like AML are applied to them. Whether they're compliant is a different story. And internationally, there's a huge amount of lack of regulation.

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I think what $100,000 or less is on the blockchain labeled with Hamas, and all of a sudden terrorist financing because there was an event and because it was visible. And I think the problem with Coinbase is that banking for so long had the excuse of invisibility that our system, as applied to a bank. Like when you compare the AML structures of an HSBC or the sanctions, when you're talking about shell companies and how hard it is for to see. And the problem is, it is so much harder when it's so visible.

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They are literally victims of their own success in the sense that it's so easy and you can see it, but then when you can see it, you have no excuse and no ability to say, like, oh, it was so difficult to see. So I think we just have to be careful when we're. There are aspects of tradfi and crypto that are so different that it's a little bit like. And I always say it's like comparing apples to shoes. And the opaqueness of banks gave it this plausible deniability that, oh, we're doing it so well, it's like, you weren't doing it well. Authorities didn't bring enforcement actions as much as they should.

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everybody should look back and say, what happened here? Why didn't we catch it earlier? I think there needs to be something better than that. If that kind of is to bring institutional adoption. I don't think we can withstand another FTX in my view. We have to be very careful.

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It's actually a good point. Talking about we couldn't afford another FTX, I just wanted to bring it a little bit back to transparency. I know we probably touched on interoperability a little bit, but going down a few layers down to the core technology, interoperability itself is a huge need and opportunity and problem that we have in blockchain. But it's also, I just thought it was something to really nail in a massive risk and also problem area of concern. Whenever we're looking at a project that as soon as a bridge is mentioned or as soon as the tech stack gets very large and you're depending on and inheriting other technologies and other companies infrastructure, unfortunately, this is necessary. I shouldn't say unfortunately. It's the way that the industry moves forward.

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But generally, when we think about soundness, safety and soundness regimes, those are the things that make you a safe business to deal with for consumers and investors, and a sound business for investors to deal with for your counterparties. What is soundness for DeFi applications? I'll throw out some prompts. It can be things like capital adequacy, it can be asset quality, it can be earnings and profitability for something that looks like a good investment opportunity. It's liquidity, operational safety and resilience, and cybersecurity.

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Part of being a sound business is not just that you operate well as a business, but it also means that you can either resist or then adapt when faced with systemic shocks to make sure that you can recover in a good and effective way. A lot of them are things that pertain to market integrity, that pertain to its efficacy of a business, massive manipulation and conflicts of interest and fraud and cybersecurity issues.

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How sound is Defi? And how do we assess and improve soundness in TradFi? So if anybody here has background TradFi, like, thinking about the soundness principles, what can we apply in thinking about the soundness of dapps and defi? Because in the point that you made about FTX and that we can't, I guess this points a little bit back to the transparency discussion. There's been lots of open source reporting about the illicit finance in this space. Some of that is because of the state of the regtech that have existed for many years in highlighting this stuff, but also because we kind of understand the idea of illicit flows. Like once we see a crypto address has been associated with ransomware.

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Has there been a lot of public reporting on soundness failures? Was there a lot of reporting on FTX being a giant fraud before it went down? Seems to catch a lot of people by surprise. So this goes back to that transparency issue. Do we have the transparency that we need to assess

soundness, not just the illicit finance stuff? Because I'm with you that the transparency problem on illicit finance, it cuts both ways with the industry, but then with soundness, do we have the transparency that we need? And is that the lack of available data or just the lack of reg tech and people actually using it to understand the space?

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There were people who were actually sounding this, usually because what they were saying didn't make any sense. And anybody who had been in diligence or non fraudulent operations knew that what they were doing was crazy. It was a fraud problem that happened to be in crypto. And I think that's like a bigger discussion.

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Yeah, I mean, FTX is a different issue, but what I want to talk about is kind of. I think you were asking more about the availability of data for transparency so that you can ensure soundness. I think there is an issue not only with the availability of data, but on the mechanism by which you assess at different points in time or even in real time, these assets and these services that you are providing on top, the traditional way of doing it every quarter by the financial statements and the transparency of the company no longer applies in this world that we are living now. You need to have, like, continuous evaluation.

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Continuous valuation, you need to have triggers on your risk framework where an assessment might be triggered because something went outside the parameters that your risk people were comfortable with. And this is something that you will see doesn't exist today in traditional finance, in any financial services that assesses risk, even kind of insurance companies, risk rating companies, they don't have the necessary infrastructure to do that. It's a completely different way. And I mean, it pushes the boundary of the new kind of tools that you need to be doing your job at the same time. And I'm going to tie that with AI.

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But the thing is, as data becomes more, the volume of data becomes overwhelming. You cannot expect a risk analyst on those frameworks to be able to understand what all these key risk indicators mean. What are the right thresholds? So you need to start thinking about copilots and about risk copilots that can help the risk teams scale. So we are talking about, I think, two things. Where there is the real time element, there is application of AI as a copilot, and all of that needs to help to ensure the soundness. Sometimes I mentioned this, that you need like a risk firewall, similar to the information age where you set up your firewall, it has the rules, and then it knows how information flows in and out of an enterprise. Similarly, you need to set up a risk firewall in financial institutions.

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You can no longer be doing it manually every quarter and saying, this is approved for the next quarter. Let's look at it kind of a quarter down the road. So that needs to fundamentally change if digital asset value is going to be flowing in and out in a live way. Do you mind if I jump in as well? Of course, if that's okay. I find this a really interesting conversation, because at the core of the blockchain industry, you would hope that soundness itself is a very achievable goal, and that is, technically speaking, almost an embedded feature. But then, yeah, once. So that's the ideal in reality, again, with the interoperability of the larger ecosystem as a system, as well as talking about centralized, moving into decentralized sources of funding and infrastructure moving back out, and then look at ulterior motives.

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We mentioned privacy tokens and things of that nature that do create difficulties here in terms of soundness and audibility, but also regulation. And where do we draw the line as to what is allowed from an infrastructure perspective, from a business perspective, and the businesses that get funding or the businesses that get support and adoption, what gets supported there by the industry as well? So you would expect that sound to self safety, long term success, the ability for a business to be predictable and to thrive over a long term, for funds to move fluidly, you'd expect for all these things to be quite easily achievable. But yeah, where we're at in the industry, it's almost the opposite.

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And most blockchain projects or companies that start have a massive revenue problem and they have a massive cash flow and profitability problem. I'm sure everyone could agree the amount of web3 or blockchain enterprises even just whatever it is, whatever sort of business model that gets funding. There's only a few business models in the blockchain industry that actually know nice balance sheets and income statements. I just find it a really interesting conversation because you would expect that this would be an easy. That things would work like they're supposed to, but unfortunately it's not how industry works.

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In computer science, we can define the soundness for the protocol itself. It's a very unified definition. Can we define the unified definition of the soundness for financial product? Or can we have that unified checklist to evaluate if that some financial product is sound or not? In my sense, I don't think we can create that unified checklist for the financial product. But it's a problem of a human understanding for the customer, for example, that if the customer have a complete understanding of that, what kind of risk inside that product, and the customer tolerate that product.

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When I think about safety and soundness regulation, the objective of that traditionally in financial services is not to make sure that the end of it is not to make sure that the company itself is sound. It's not to make sure that the shareholders don't lose value. Because that's essentially, I mean, your shareholders are there to take risk and you're supposed to appoint certain people to exercise certain sort of degrees of governance over the company. You're supposed to be part of that governance process. Right. The point of safety and soundness regulation, and where safety and soundness matters is are you going to lose a customer's money? Are you going to be able to repay your liabilities? If somebody else is placing their trust in you, are you making good on that?

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And so when I think about things from a DeFi perspective, actually a lot of these deFi services seem to me to be based on a thoroughly financially conservative premise. Transactions are over collateralized, there's automatic triggers in place for liquidation and so on and so forth. And so from that perspective, safety and soundness is not immediately what comes to mind. When I think about safety and soundness, financial risk is not really what comes to mind. What comes to mind is, are you going to lose your customers money because somebody hacked you and stole all the funds? To me, that is sort of like the core of safety and soundness. Isn't that security? Because I feel like when people, and do those terms get conflated?

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And I'm not saying that's not right, I'm just saying, am I thinking about it wrong that whenever I hear people talk about that and I hear security, is security part of safety and soundness? Or if it's not in traditional finance, should it be considered as such, given the hackability of DeFi, where is the soundness problem? Or is the safety and soundness problem in DeFi security? Nomenclature point. In financial regulation, when you think about different buckets of regulation, you've got sort of like prudential regulation, right? It's one big pillar. And traditionally, a lot of that is around financial resources as well as management of risk on the balance sheet, right.

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Depending on what type of business you're doing. So, for instance, if you're running a bank, then you want to make sure that your balance sheet is sufficiently sound. You've got liquidity risk taken care of, and a lot of that is financial in nature or any kind of institution. You want to make sure you've got enough financial resources that are liquid so that if you do go down, there is enough in place for an orderly windup. So operational risk is typically seen in sort of like a separate bucket. And that's where you put your cyber risk. It's where you put all those other things that can go wrong.

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I think in the case of crypto in general and DeFi in particular, we are seeing that technology getting so far embedded into the business model that essentially these protocols live and die by, whether they are cybercrime is the financial crime when you're targeting DeFi. So the ability to repay your liabilities is affected by. And this is where safety and soundness obligations. And I guess in my conversations, it was always sort of viewed as a contributor to your ability to be able to fulfill your debts to consumers. Safety and soundness regimes make sense from a business investment perspective. There's a financial incentive for you to be good at cybersecurity, because whether you're TradFi or DeFi, that will keep you alive as a business.

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So safety and soundness stuff, at least, is positive, because, golly, at least we could say, if you are a sound business, you will stay alive as a business. But you're right, they're not good at it. And inherently, when you're targeting DeFi, it is a financial crime, because none of these things are being targeted to not steal the money or to compromise it in that way. So this issue of cybersecurity, and I really appreciate dissecting the issues of soundness. Part of what I want to prompt and continue this conversation is asking, are there specific features of DeFi that affect this soundness of it? The fact that it's entirely software based, like the assets and its operations, mean that it has a very digital attack surface, other features about it that affect the soundness and then the so what do we do about it?

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So, thinking about soundness frameworks that we have, like what can and should be done to improve the soundness in these ecosystems. Things like code audits and stuff have come up before, but I'll throw that out to the group to bring up the discussion. I guess that's what I was getting at, was the idea that right now, in traditional finance, going back to what you said, there's not this analysis. Well, maybe there is in my, because I have my life in tradfi, nonexistent. So I guess my question is, where does the analysis of security of the bank, the ability to get into the safe deposit, the safe, because the risk of theft or their cybersecurity. Right. Like if somebody's going to hack into a bank with DeFi, it is all in the code.

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The idea that you don't have code audits would literally be like somebody insuring a bank that had no lock on its front doors. And the idea is so wild, the idea that you don't have a safe, you don't have locks on your doors. Like how could this be allowed? So what is the comparable thing that you would have in DeFi? Presumably a code audit. So how is that not like baseline requirement in the sense that if you want safety and soundness in Defi, if that's where it goes, you have to have something that literally puts a lock on a door. Otherwise you just have millions, if not billions of dollars going into a vaultless, lockless, securityless building?

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I can answer that. So what you'll see from financial institutions, you will see right now two fundamental risk frameworks. The one is about onboarding assets and offboarding assets, where they want to make sure that all the representations that are being made on those assets remain as is. And they know everything about those assets if something changes. And they care a lot about even kind of social media sentiment and the token price and all of that. Right. And level of decentralization, things that they need to know. There is also kind of the custody frameworks, which I think go to your point, which is. So it's a different thing. You onboard the asset, you might have ETH that you are trading on, but you might not be kind of safekeeping it for the users.

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So trading has different requirements, but you do make a representation to your customers that you can trade on ETH. So to some extent you have gone through what ETH means and kind of all the parameters. Now, if you're doing custody, you need to ensure the safekeeping of that asset and that asset doesn't go away. So you have a different framework now where you look at the smart contract, you look at what kind of compiler on the smart contract, the structure, who holds the keys. You use blockchain analysis to figure out which entities get attached to that, you understand the transactions in and out, you understand all of that. At the same time, you bring key risk indicators from your internal custodian solution.

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And I second everything that's been said in this part of the topic so far. One thing that I want to just point out as well is we tend to go straight to hacks in Defi. Can funds be stolen? One of the things around smart contracts and blockchain infrastructure is that obviously it's generally permanent. And it's not only hacks that we're worried about. Some of the worst things that can happen are optimization issues and long term unpredictable effects of multiple transactions in a row that were in the wrong order. That wouldn't happen usually. So if a protocol was deployed with only 98% test coverage. So most mathematical possible scenarios were tested. So you can have these essentially businesses that are like a decentralized exchange that's having, quote unquote revenue and it's making heaven transactions.

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And then after a while, you start to see these small optimization things that, although it's not potentially one hacker going back to soundness, users end up losing money. Or for example, the gas optimization sucks after a while, or in certain scenarios that weren't thought of when the smart contracts were deployed. And so, yeah, users start to lose money. Maybe the investors lose money because that one optimization issue causes things. So I just want to say, you shouldn't always go directly to the worst case scenario being a hack. And a comment was made about requirements of audits. It is interesting that in every other industry, if you build a skyscraper,

you're expected to have some third party checks. But if you build a protocol with billions of dollars of users money, you can theoretically just press deploy, and you're good to go without any checks.

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Yeah. Interestingly enough, I think now with tokenization of real world assets, there is a lot of interest by financial institutions in bridges. And they want to apply some of the bridges in terms of the private or hybrid protocols where they're running their experiments or production. I think that becomes extremely interesting, because now you want to use bridges and you want to apply the risk frameworks that financial institutions feel comfortable on those bridges. I think that those kind of use cases bring some of the elements of the public blockchain world closer to the financial institutions and put them under scrutiny in terms of the risk they're taking. And this is kind of quite interesting, right, because it's still driven by the token price.

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Because if you're a bridge and you get endorsed by one of the largest financial institutions and you index yourself towards tokenization, your price kind of goes up. So we see a lot of players, like being willing to comply and figure out what does it mean? What kind of data do we need to make available? How do we make financial institutions feel comfortable using us so that they can drive volumes on their bridges and kind of be more successful and see more adoption?
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Like when there's a bug in your software, you now have to get everybody to agree to pushing that software fix and that update, whether it's an optimization issue or whether it's a software patch that's patching a vulnerability that we recognize that Lazarus in North Korea is about to hack for half a billion dollars either way, that's one of the again, that immutability and censorship resistance, combined with the decentralization and absence of a clear line of accountability, make the fix and the way forward on soundness interesting.

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The proposal of an DeFi paper was a voluntary opt in approach on partnership for thinking about underlying infrastructure as basically using the same framework that we do for critical infrastructure and imposing that on DeFi. My position is that they already are critical infrastructure, and it's not, in fact, their decision to opt in. They are covered. It's on SSRN, and there's a link to it in the discord. But thoughts from the group about what is needed to improve soundness in these ecosystems. We talked about code screening and audits. What are the other kinds of mechanisms that we should probably be thinking about for improving soundness inside of diva ecosystems. We talked about better calibrating risk, understanding risk, mapping out the key questions to make sure that we can map it to the specific data needs.

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And is that information either publicly transparent and if not transparent, through some other mechanism? What else from the group do we think would benefit to improving the soundness and ensuring the soundness of Defi?

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If I can start around, of course, I think education is a very big component, and I think education can go across the entire ecosystem, from simple users that participate in those projects, to the regulators, to financial institutions. I think this is so new and so complex that we need to make sure that we create. I think there need to be separate companies that dedicate themselves on how to educate people in this space. So I think it's a big barrier. And particularly given that we're living in an era of AI and people have to deal with the AI risk as well, it becomes extremely. This bucket of emerging risk that they call is becoming more and more complex. So I don't think people have the resources on the other side.

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You have a community that's enthusiastic and nobody's telling them otherwise, and nobody's telling them do that at your own risk. So I think somehow there needs to be some kind of educational component there. Potentially I can follow up from that. Go ahead. I think that's a very good point. One thing I worry about is the business models that are within the blockchain industry. There's a lot of short term business models, and some of the short term market dependent business models make a lot of. They're very appealing and they make a lot of short term revenue. There isn't a lot of longer term business models. Obviously, we're in emerging tech environment, so longer term business models are harder.

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But most blockchain companies have a revenue and have a financial problem, which I do think leads to less security procedures, less of a quality of a board and the team. And I just think most of this stuff comes down to how much funding they have. And most of their funding oftentimes goes to marketing, development, and not to the procedures that will result in soundness. So I think the actual business models hopefully mature over time within the blockchain industry. And I think that's probably one of the main things I think about.

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I think we are dealing with an ecosystem full of players that are, shall we say, not sort of necessarily rooted or used to thinking in a risk mindset. With a risk mindset, it is an, At the same time, and. And it's an. It's an area where, you know, the applicability, in a practical sense of regulatory frameworks is still evolving. And so I think that means two things. I think the first thing is that we need to start somewhere and not let sort of the perfect be the enemy of the good.

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That means pinpointing, not necessarily trying to develop a whole full scale framework of what do we think Dapps and deFi services should be doing, but identifying what we think are the most life or death risks when it comes to these sorts of entities. It's not just hacks, it's also other sorts of code vulnerabilities, inadequate testing and so on and so forth. But I think that is probably the key area alongside AML/CFT that we need to tackle. And I think it is really important for the industry and for groups like begin to drive conversations on what those standards look like.

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If you leave it up to sort of regulators to decide exactly what should be done, they will tend to be an overreaction after a crisis. And so it is much better for the industry to work out what is practical ahead of time and what is reasonable and meaningful and to have that in place and to have players that are trying to do good, sign up to meet those standards than to have all of this try to be sort of pushed down the regulatory pipeline.

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I'm going to use FDIC insurance as an example that doesn't fall under the concept of safety and soundness. Yes, FDIC would prefer to insured a business that's sound, I think. So some of their obligations under their safety and soundness regime, basically, you need to meet those in order for them to be willing to say yes. And now you get insurance. Okay, so I wasn't far off. So that's great. So I guess here's my question. There's a discussion in the industry about the need for an SRL, like a self regulatory organization. Right. And that's a very difficult conversation given, kind of like where some of the industry members are. But you could have self insurance in the sense of if the FDIC isn't willing to do this, you could have protocols that say, oh, we're going to self insure, we're going to offer this.

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A lot of this money does go to marketing and other things, but they could actually tag some of it first to say, as a baseline matter, we're going to basically say that we're going to insure up to $5,000 of loss or some amount of loss. That's like, if you put your money in this, we guarantee that you won't lose this amount. And the industry could do that if it wanted to show that it cared about that. Right?

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And I don't know, that's a crazy idea, but you could certainly propose it and say, look, if there's a protocol out there that really cares and maybe the hack is not your fault, maybe you get an audit, maybe there's an optimization problem that you didn't anticipate, but then you self insured for this to protect your customers in the gap, that industry, that regulation hasn't filled. Is that

wildly? I mean, I know it's maybe not probable, but is it an unreasonable suggestion? Are there any insurance companies that say, yeah, I will insure you if you get a good audit?

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That is a great question. And it's definitely something that auditing firms have. It's a natural thing that we've looked into. It becomes difficult because then they try to put you into different categories. The obvious one, cyber insurance. That's the obvious one. But there's a few nuances into why that often doesn't work. It goes down to the specific actual underwriter and then the chain of where the funds come from. Put it shortly. There isn't too many established companies that are comfortable. There's a few companies that market themselves as like blockchain insurance companies. But the problem is, when you really go down the rabbit hole, they're often getting there if they are properly regular insurance company.

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Yeah, sorry, just in our experience. Anyways, I've seen a couple of examples of insurance but they're mostly around centralized providers of services. They had some company that would ensure you don't lose the stake rewards and they would take that risk. There is, I think, a company called one degree and they offer to custodians and they go to regulators and they say, can we get this as a requirement for those that get digital asset licenses? But for issuers of tokens, there is no incentive for them to do it because they can go elsewhere in the world and kind of float their token and within two or three or four years they're done. So why spend time, like on going through that?